State-of-the-Art Overview of 3GPP Ambient IoT Radio Access Aspects

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Abstract

The number of connected IoT devices is expected to reach billions in the foreseeable future. The prosperity of the IoT market attracts the interest of fifth generation of cellular network (5G) and beyond evolution to promote new IoT technology supporting passive devices. Such devices, which are battery-less or with limited energy storage, are powered by harvesting energy from the ambient environment and are referred to as Ambient IoT (A-IoT) devices. This paper provides a state-of-the-art overview of the third generation partnership project (3GPP) A-IoT radio access techniques. We describe use cases, scenarios, and distinct device architecture and its impact on designs. We discuss the challenges and solutions of A-IoT and demonstrate advancements over existing passive IoT system in terms of link performance, coverage, and inventory efficiency.

Introduction

The third generation partnership project (3GPP) has designated Long Term Evolution for machines and narrowband Internet of Things (NB-IoT) as low power and wide area (LPWA) solutions. The development of the fifth generation of cellular network (5G) new radio (NR) brings new capabilities to those solutions [1]. Further advancements of Reduced Capability (RedCap) add a new dimension to IoT connectivity [2], offering higher peak data rate, lower latency, and improved power efficiency to IoT terminals. However, the existing designs cater to active terminals powered by battery. To achieve ubiquitous connectivity in the future, new IoT technologies supporting passive terminals being battery-less or with limited energy storage and powered by harvesting energy from the ambient environment are required. The existing passive communication technologies, exemplified by radio-frequency identification (RFID) [3], are launched for affordability and ease of deployment. However, it has a shortage in communication range which makes it difficult to fulfill demands in wide areas and seamless coverage. To bridge the gap, 3GPP has initiated a study item (SI) for Ambient IoT (A-IoT) in NR Release 18 (Rel-18) [4]. The study defines key design targets and investigates representative use cases, deployment scenarios, device types, and connectivity topologies. Based on recommendations from the study, 3GPP has further launched a SI in working group in Rel-19, aiming to comprehensively assess the feasibility of the A-IoT solutions and to investigate detailed radio access techniques to gain a competitive edge over other passive IoT technologies. A-IoT intends to address use cases and requirements that cannot be satisfied by the existing 3GPP LPWA but not to replace it.

The main contribution of this paper is to provide a state-of-the-art overview of 3GPP A-IoT radio access aspects. The paper first introduces use cases and scenarios and highlights the device architecture that has impact on designs. Technical challenges and solutions are then discussed. Various performance analysis is provided to verify that A-IoT outperforms traditional passive IoT technologies. Next, the paper describes the overall system design. Finally, the paper provides an outlook on future research and applications.

Use Cases and Scenarios

The Rel-19 A-IoT study addresses the following use cases and scenarios [5]:

1. The study focuses on two representative use cases, i.e. indoor inventory and indoor command.
2. Two combinations of deployment scenarios and topologies are investigated. For devices with Device-to-Reader (D2R) backscattering on an external carrier wave (CW), CW inside and outside topology (bi-static backscattering, -B case) are considered for both scenarios. For CW inside topology, both bi-static (-A1 case) and monostatic (-A2 case) backscattering are considered, as shown in Figure 1.

- Deployment scenario 1 with Topology 1 (D1T1), where D1 refers to device and base station (BS) indoors, and T1 refers to device directly and bidirectionally communicates with a BS.
- Deployment scenario 2 with Topology 2 (D2T2), where D2 refers to device indoors and BS outdoors, and T2 refers to device communicates bidirectionally with an intermediate node between the device and

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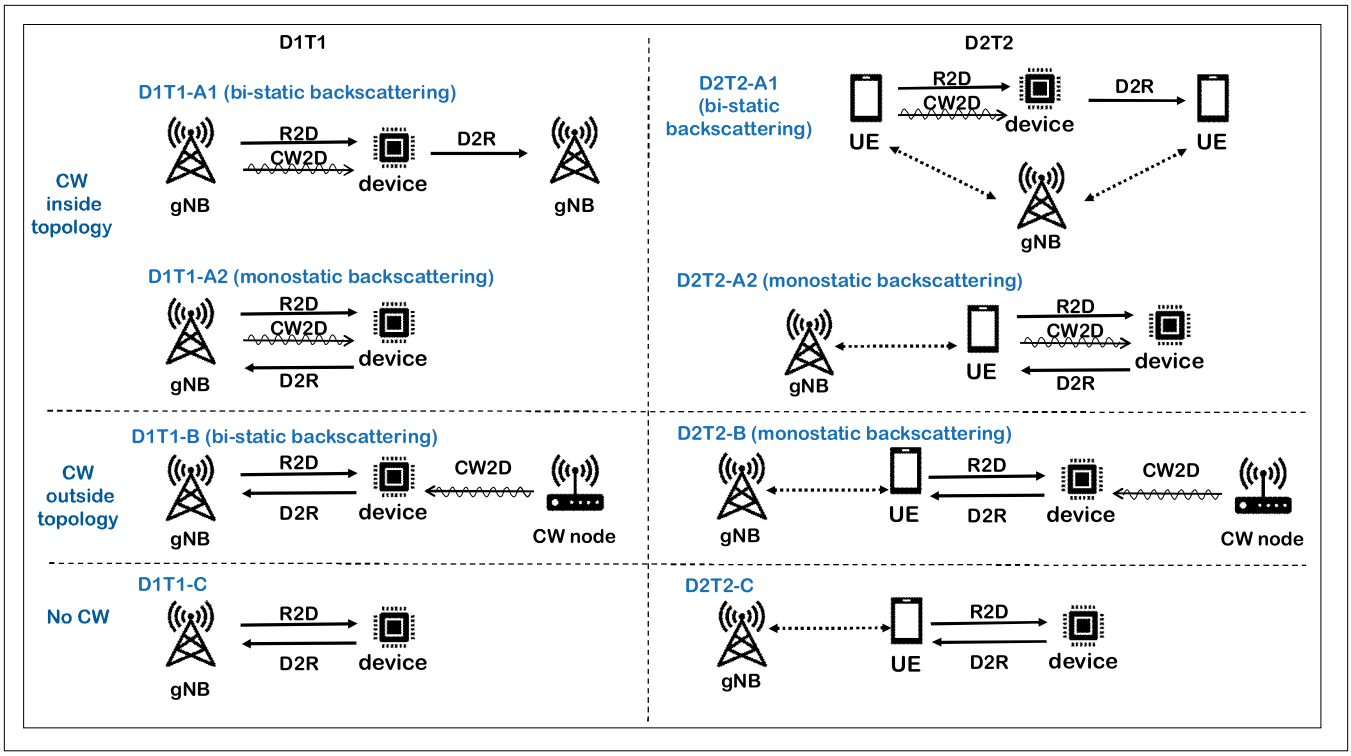


Figure 1. Illustration of deployment scenarios, topologies and CW transmission cases.

outdoor BS, where the intermediate node is a user equipment (UE) under network control.

Device Architecture

Three A-IoT device types are classified based on power consumption and D2R transmission schemes.

- Device 1: It consumes ~1 μW peak power as neither Reader-to-Device (R2D) nor D2R link amplification being performed, has energy storage, the initial sampling frequency offset (SFO) is up to 10^5 ppm, and D2R transmission is backscattered on an external CW.
- Device 2: It consumes a few hundred μW peak power with either or both R2D and D2R link amplification being performed, has energy storage, the initial SFO is up to 10^4 ppm. Depending on how D2R is transmitted, device 2 is categorized into:
  - Device 2a: D2R transmission is backscattered on an external CW.
  - Device 2b: CW of the D2R transmission is generated internally.

Device architecture consists of antenna, energy storage, receiver components, transmission blocks, etc., details refer to [5]. Device architecture blocks impacting designs are presented.

- Clock for Sampling and SFO. The clock for sampling is an on-chip resistance capacitance oscillator, with an MHz level sampling frequency. With ~ 1 μW power consumption, it is challenging to calibrate the clock frequency. Counting the sampling points can provide clock calibration but the device SFO is still 10^4 ~10^5 ppm.
- Capacitor: All device types have energy storage capability by implementing μF-level off-chip capacitors. Compared to RFID tags with only pF-level on-chip capacitors, energy storage capability of A-IoT devices is beneficial to reduce the activation threshold and enlarge the coverage.
- Digital Baseband (BB) for Small Frequency Shift (SFS). The digital BB includes functional blocks such as an encoder, which can enable SFS by repetition of codewords, or modulation codewords with a square wave.
- Clock for Time Counting. Once a device performs the transmission/reception (Tx/Rx) procedure and runs the clock for sampling, the power consumption is much larger than that harvested from the environment. A clock with lower frequency of tens of kHz consumes less power (e.g., ~ 0.1 μW). It is reported that such a clock can be used to wake up device from SLEEP state [6], allowing the device to conserve and harvest energy to extend its available time.
- Local Oscillator (LO) and Carrier Frequency Offset (CFO). For device 2b, an LO is required for up/down conversion of carrier frequency. The CFO after calibration within 10s ~ 200 ppm is necessary to reduce the guard band of the D2R transmission. To achieve so, an R2D signal to assist CFO calibration may be useful.

Technical Challenges and Solutions

Coverage Aspects

A-IoT aims to reach a larger coverage of more than 10 m [5]. Techniques to improve link performance are investigated.

Timing Recovery Under Large Device SFO

The RFID reader is close to tag, even with a large device SFO, the self-clocking line code

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can facilitate the D2R timing acquisition and non-coherent edge detection; however, it is not applicable for A-IoT with larger coverage and weaker signal strength. Hence, SFO correction is needed for the reader to implement advanced processing. A precise SFO estimation approach based on D2R preamble and midamble/postamble at reader side is proposed:

- Step 1: The reader roughly estimates SFO by performing a certain number of SFO hypotheses based on the preamble.
- Step 2: With the rough SFO, the reader determines the search window of midamble/postamble detection.
- Step 3: The fine SFO is derived by the indices of peak timing correlation obtained by both preamble and midamble/postamble.

Assuming a device SFO of 10^5 ppm, the proposed method achieves a residual SFO less than 0.03% [7] and is sufficient for the reader performing coherent detection [see lines with square marks in Figure 2(a)].

Forward Error Correction

The forward error correction (FEC) code is introduced to improve link performance considering the limited device complexity. For R2D, typical FEC codes require high decoding complexity that is not capable for device 1. For D2R, encoding a small payload and by limited number of registers is feasible. A convolutional encoder consists of *K*-stage shift registers and *n* modulo-2 adders (output generators) is easier for device to implement and provide sufficient link performance gain. Compared to no FEC, more than 4 dB gain at 10% block error rate (BLER) is achieved using FEC, assuming the same chip rate [8].

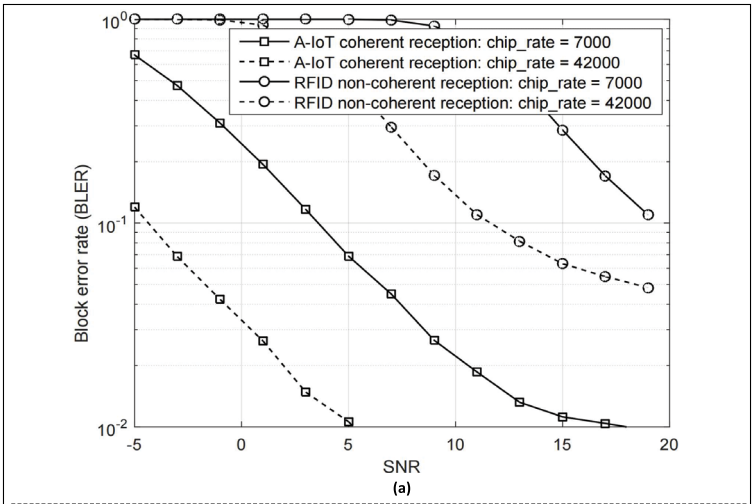
Carrier-Wave Waveform

For D2R backscattering on an external CW, a single unmodulated sinusoid tone (waveform 1) without frequency hopping (FH) is considered as baseline. Waveform 1 with FH and two unmodulated tones (waveform 2) are further considered to improve link performance.

Benefit from frequency diversity, both waveform 1 with FH and waveform 2 provide performance gain. Simulation shows that compared to waveform 1 without FH, waveform 2 with 7 MHz gap (30 ns delay spread) provides 1.2 dB and 5.8 dB gain for 10% and 1% BLER, respectively [9]. Similar results are achieved using waveform 1 with FH. The gain depends on the gap between two tones or two hops, corresponding to the coherence bandwidth. The impact on device is trivial since backscattering is unaware of multiple tones or hops. Additional complexity may be introduced to CW node for CW generation and to reader for D2R data combination and CW interference suppression.

Coverage Performance

For D2R, link-level simulation is performed to acquire the required signal to interference plus noise ratio (SINR), and link budget calculation is then performed to obtain the coverage metric maximum path loss [5]. Figure 2(a) compares the D2R link performance of coherent detection and non-coherent detection (i.e., typically used in RFID). It is observed that with the proposed enhancements including precise SFO estimation and FEC, the target 10% BLER is achieved at -4.5 dB, assuming 42k chip rate. With the same chip rate, more than 15 dB is obtained when compared to the non-coherent receiver.



(b)

Device	D1T1 (BS reader, non-line-of-sight scenario)				D2T2 (UE reader, line-of-sight scenario)			
	R2D distance	D2R distance	Coverage (bottleneck link distance)	cp. RFID	R2D distance	D2R distance	Coverage (bottleneck link distance)	cp. RFID
RFID tag	2.35 m	8 m	2.35m	1x	2.41 m	3.6 m	2.41m	1x
Device 1	36 m	-A1/B: 135 m	-A1/B: 36m	15x	13 m	-A1/B: 135 m	13m	5x
		-A2: 24 m	-A2: 24m	10x		-A2: 24 m	13m	5x
Device 2a	95 m	-A1/B: >200m	-A1/B: 95m	40x	45 m	-A1/B: >200m	45m	19x
		-A2: 46m	-A2: 46m	20x		-A2: 34m	34m	14x
Device 2b	95 m	>200 m	95 m	40x	45 m	>200 m	45m	19x

Note 1: For RFID, the activation threshold for R2D reception is -23 dBm and non-coherent receiver is applied for D2R reception; for A-IoT, the activation threshold for R2D reception is -36 dBm for device 1 and -45 dBm for device 2a/2b, and coherent receiver is applied for D2R reception.

Note 2: For A-IoT devices, for D2R backscattering, -A1 case and -B case is bi-static backscattering; -A2 case is monostatic backscattering. Details refer to Figure 1.

Figure 2. a) D2R link level performance. b) Coverage evaluation results.

For link budget evaluation, additional aspects are considered for D2R transmission:

- By switching between different impedance states, a backscatter loss due to modulation factor exists, which quantifies how much the backscatter device can alter the reflected signal. Improving the modulation factor is crucial for enhancing coverage [10]. Theoretically, the backscatter loss is 0 dB for binary phase shift keying (BPSK) and 6 dB for on-off keying (OOK).
- On-object antenna gain penalty is caused by object attachment that reduces the antenna radiation efficiency and distorts the antenna pattern. A typical value of 0.9 dB and 4.7 dB for materials such as cardboard sheet and pine plywood are respectively considered [10].
- The CW interference cancellation capability depends on several aspects including spatial isolation for separate Tx/Rx antenna configuration, analog cancellation from antenna to analog-to-digital converter, and digital cancellation at baseband. For monostatic backscatter (i.e., -A2 case), 10-30 dB suppression is expected for radio frequency (RF) analog interference cancellation; around 40 dB suppression is provided by a circulator/directional coupler and/or the antenna with high isolation between Tx and Rx; up to 80 dB digital domain

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suppression is achieved by filtering and/or reconstructing-then-subtracting the interference from the received signal. Filtering can be achieved in frequency domain or spatial domain. Frequency domain filtering depends on the characteristics of the CW waveform, and spatial domain filtering can be fulfilled by the classic minimum mean squared error-interference rejection combining method. The implementation of reconstructing-then-subtracting interference is more complex, as it requires accurate CW waveform modeling and channel estimation [9].

For R2D, rather than using the required SINR to calculate the receiver sensitivity, a device with RF envelope detector utilizes an activation threshold as the receiver sensitivity. Assisted by the energy storage capability of device, the receiver sensitivity becomes lower than an RFID tag. The value is reduced from −30 dBm to −36 dBm for device 1 and from −40 dBm to −45 dBm for device 2 [5].

Coverage results are provided in Figure 2(b) [11], according to the methodology in [5]. Taking device 1 under D1T1 scenario as an example, it is observed that a promising coverage result of more than 20 m is achieved. The bottleneck link for bi-static backscattering (i.e., D1T1-A1/B) is R2D as it is limited by the R2D receiver sensitivity; while for monostatic backscattering (i.e., D1T1-A2), subject to the CW self-interference cancelation capability, D2R is more relevant to the overall coverage. In addition, comparison with RFID is also shown. The coverage of A-IoT significantly improves under all deployments. For scenarios where R2D is the bottleneck, since A-IoT has a much lower device activation threshold than RFID due to energy storage, the coverage is improved more than 15 times. For scenarios where D2R is the bottleneck, with the proposed enhancements, required SINR of A-IoT reduces approximately 10 dB.

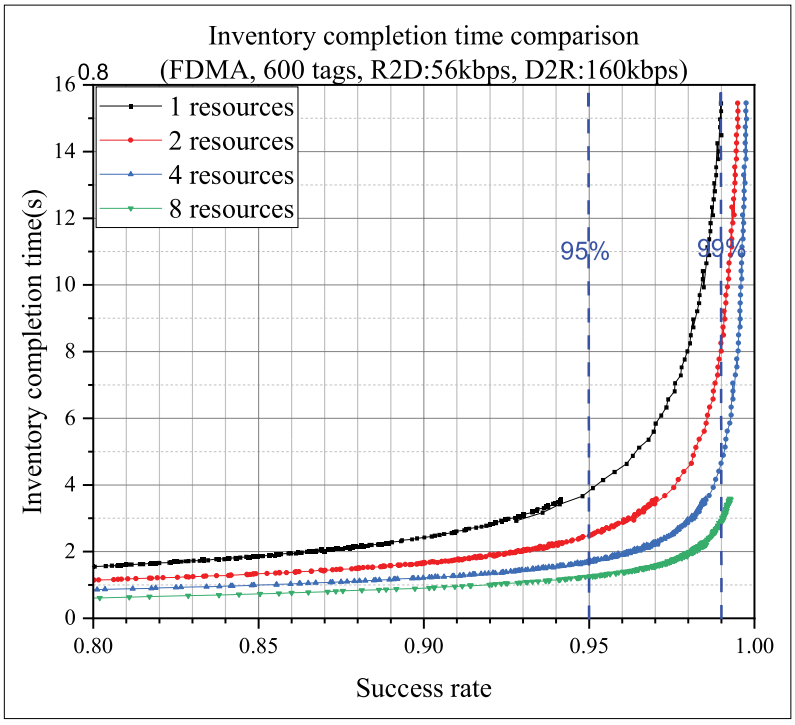


Figure 3. Inventory completion time using FDMA.

Inventory Efficiency Aspects

Inventory efficiency, measured by metrics such as inventory completion time and the percentage of utilized slots, is a critical design factor.

Harmonic Avoidance Frequency Division Multiple Access

Slotted ALOHA is adopted in the existing passive IoT system, of which the theoretical maximum percentage of utilized slots is 36.8%. Within each time slot, allowing multiple devices to transmit D2R in frequency division multiple access (FDMA) provides higher inventory efficiency. Figure 3 shows that the inventory completion time is significantly reduced by 46.5%, 69.1%, and 80.6%, respectively, assuming 2, 4, and 8 FDMA resources, if 99% of 600 devices complete inventory.

FDMA is achieved by applying SFS through baseband operations, although the power consumption and complexity of a device is quite limited. Either repeating a codeword R times (R is SFS factor) within the duration of an information bit or modulating a codeword with a square wave (R periods within an information bit) leads to peak lobe of the spectrum shifting in frequency domain.

Harmonic interferences are more severe due to the OOK modulation. Theoretically, the power of the 3rd and 5th harmonics is approximately -9.5 dB and -14 dB, respectively, compared to the main lobe. In addition, with expanding coverage in A-IoT, devices accessing using FDMA may suffer from severe near-far effect. Therefore, the SFS factor R should be carefully selected to avoid inter-device harmonic interference.

Anti-Collision Access Strategy

Besides optimizing FDMA under the framework of slotted ALOHA, other anti-collision access schemes are investigated to further improve access efficiency. Bi-tree algorithm and its variants are promising schemes. Bi-tree algorithms are deterministic, meaning that all devices can be identified within a given time. When compared to slotted ALOHA with large portion of empty slots, some bi-tree variants can reduce empty slots and significantly improve efficiency to 50% [12]. One issue of bi-tree is the higher device energy consumption as it usually requires a device to respond continuously before being identified. To overcome this drawback, a hybrid algorithm combining slotted ALOHA and bi-tree is proposed [12], which potentially improves the efficiency to 65%. In the proposed algorithm, the slotted ALOHA first divides all selected devices into multiple subsets, each containing a few devices. The bi-tree algorithm then identifies devices within each subset. Due to the decrease in collision domain, devices respond less frequently before being identified. Further improvement in efficiency is pursued considering more suitable grouping and other technologies.

Constrained Energy Aspects

Unlike RFID, A-IoT devices can install capacitors of 1~10 μF, which can store energy so that the activation threshold is reduced, and coverage is increased. However, a farther distance along with a weaker incident signal strength, making a device

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difficult to support a charge-and-play mode, and leading to device unavailability. Improvements are required to combat the energy limits.

Energy Harvesting

For device 1 with ~1 μW power consumption, an RF energy harvest system is assumed, which is the only reliable energy source. A typical operation mode is that a device is activated after up to tens of seconds charging and performs Tx/Rx till it runs out of available energy. Therefore, a device may become unavailable for Tx/Rx due to energy harvesting. Depending on whether the device can get information from the reader on when to wake up or not, two directions are investigated:

- Direction 1: The reader does not provide any information to device, nor be aware of device (un)availability, is an implementation solution. The device operates in charge-discharge mode. The sustainable time for device 1 can be several seconds with appropriate deployments. A proposed practical operation is that the reader transmits paging messages at a level of half of the device sustainable time. Based on evaluations, all 600 devices complete inventory before running out of available energy [13].
- Direction 2: The reader can provide information to a device, based on which the device may enter SLEEP state to reserve or harvest energy. Since the power consumption of device 2 is a hundred times larger than device 1, Direction 1 is not applicable due to the heavy resource overhead for paging transmission. We propose a solution for the device being in SLEEP state before and during an inventory procedure. Evaluations demonstrate that around 30% efficiency gain on inventory completion time for 600 devices can be achieved [13].

D2R Packet Segmentation

Device availability time has some limitations to the maximum transport block size in a D2R transmission. Segmentation is a useful solution. However, segmentation in NR cannot be directly applied to A-IoT, as it requires large buffer for device. Therefore, Access Stratum (AS) layer buffer-less segmentation in A-IoT is adopted in D2R. For example, a device can record a segment by saving the starting position and the length of the transmitted data in memory without storing data in buffer. Segment indices and the total number of segments are not required, and the only necessary information is whether segmentation is applied, and the last Medium Access Control (MAC) packet data unit (PDU) is segment indication. Methods for handling data transmission failure are enhanced for segmentation. When D2R message is segmented, the reader can only trigger the (re) transmission of the failure segment(s).

Coexistence Aspects

A-IoT uses licensed NR spectrum, and designs should be compatible with NR to avoid coexistence issues.

Compatibility With NR

The designs ensure compatibility with NR in two aspects. First, the resource granularity for A-IoT transmission in frequency domain is defined as 180 kHz (i.e., 1 resource block) for R2D and as 15 kHz (i.e., 1 resource element) for D2R, which are aligned with NR resource grid. In addition, considering that OOK is the only feasible modulation and waveform candidate for devices to receive R2D, from the reader perspective, reusing the existing NR Orthogonal Frequency Division Multiplexing (OFDM) based modulator to generate OOK waveform is considered. Starting from definitions in TR 38.869 [14], OOK-4 waveform, i.e., multiple OOK chips per one OFDM symbol, is applied. An example of OOK-4 waveform generation is proposed to ensure that the OOK waveform in the time domain is shaped as a desired square wave. The transmitter up-samples the OOK chip in the time domain and then performs N'-points discrete Fourier transformation (DFT) to obtain the frequency domain signal. The frequency domain signal is mapped to the X subcarriers (i.e., the transmission bandwidth), and transferred to the time domain performing N-points inverse DFT.

Coexistence Evaluation

The coexistence analysis concentrates on a practical deployment where the NR macro-BS locates outdoor and the A-IoT reader deploys indoor. Two cases are evaluated in such a deployment: All NR UEs are in outdoor; 10% of NR UEs are in indoor while connecting to outdoor NR BS. A-IoT system occupies licensed NR Frequency Division Duplexing (FDD) 900 MHz spectrum, where the R2D transmission is in FDD DL for D1T1 and in FDD UL for D2T2. The coexistence evaluation mainly addresses backscatter communication for devices 1/2a where the D2R transmission and CW-to-Device (CW2D) link occupy the same spectrum. Both FDD DL and UL are assessed depending on whether CW node is deployed inside or outside topologies. In the evaluation, the RF Tx/Rx characteristics for A-IoT reader and CW node are additionally assumed. Upon the deployment scenarios where the A-IoT reader or CW node is a BS or UE, and the occupied spectrum, either existing NB-IoT BS requirements or UE requirements are employed.

Figure 4 depicts the feasibility of deployment scenarios categorized by potential coexistence cases [5]. A deployment scenario is recognized as feasible when victims in all coexistence cases experience negligible performance degradation. For A-IoT device or reader as victim, SINR degradation is used as the performance metric, and the degradation less than 1 dB is treated as negligible. For NR as victim, DL or UL throughput degradation is adopted, and the degradation is negligible if it is less than 5%. As shown in Figure 4, the feasible cases and scenarios are marked as green; otherwise, the questionable ones are marked as orange. It is concluded that D1T1 with all NR UEs outdoors is a promising deployment scenario for A-IoT solutions. For other deployment scenarios, potential interference mitigation may be pursued for better coexistence.

Overall System Design

The overall design of A-IoT physical layer and higher layer are briefly described.

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Deployment scenarios	Spectrum	Coexist. cases			
D1T1-A1, NR UE outdoor	R2D: DL	A	B	C	D
D1T1-A2, NR UE indoor	CW2D/D2R: DL	A	B	C	D
D1T1-B, NR UE outdoor	R2D: DL	E	F	C	D
D1T1-B, NR UE indoor	CW2D/D2R: UL	E	F	C	D
D2T2-A2, NR UE outdoor	R2D: UL	E	F	G	H
D2T2-A2, NR UE indoor	CW2D/D2R: UL	E	F	G	H
D2T2-B, NR UE outdoor	R2D: UL	A	B	G	H
D2T2-B, NR UE indoor	CW2D/D2R: DL	A	B	G	H

Coexist. cases	Aggressor	Victim
A	Device	NR DL
B	NR DL	Reader
C	Reader	NR DL
D	NR DL	Device
E	Device	NR UL
F	NR UL	Reader
G	Reader	NR UL
H	NR UL	Device

Feasible cases: (1) SINR degradation less than 1 dB of A-IoT, and (2) NR throughput degradation less than 5%
Questionable cases: (1) SINR degradation larger than 1 dB for A-IoT, and/or (2) NR throughput degradation larger than 5%

Figure 4. Coexistence evaluation results.

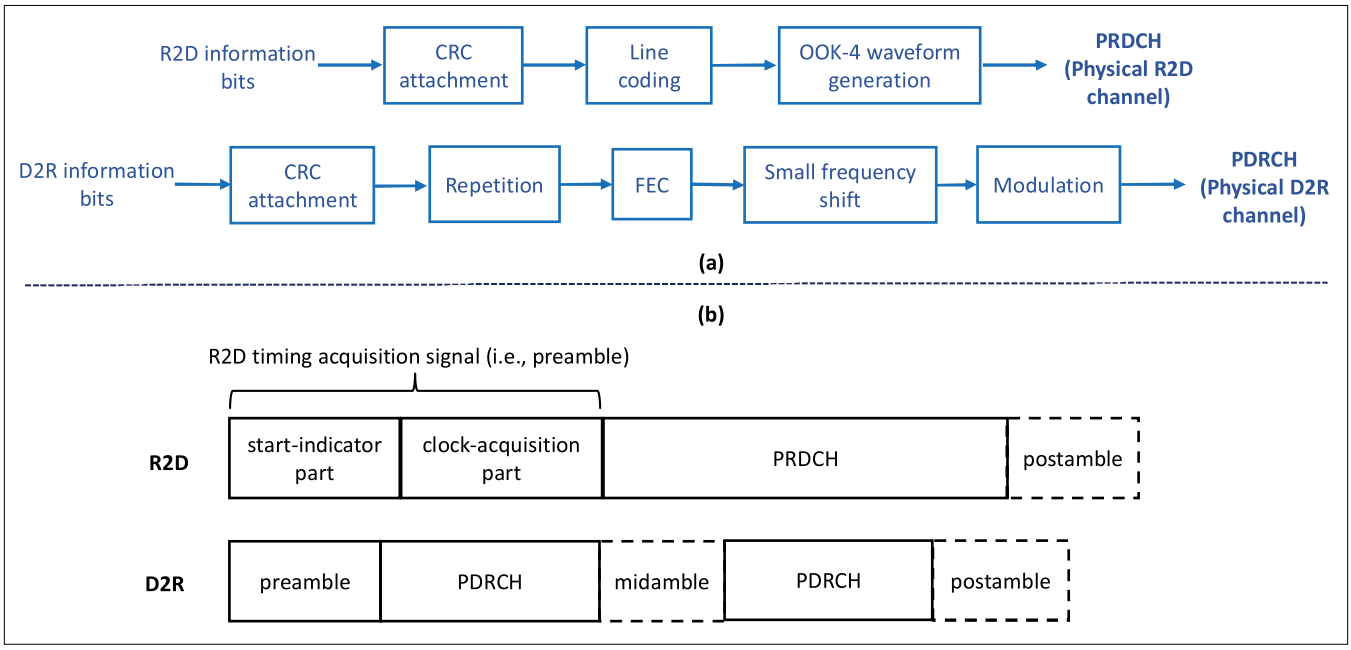


Figure 5. a) Physical channel generation flow. b) Frame structure of R2D/D2R.

General Physical Procedure

A-IoT physical designs are different from NR. First, storing large information payload, such as system information, incurs high memory cost and should be avoided. Second, as an asynchronous system, the physical channel should incorporate self-contained synchronization signals. Third, the physical channel design is simplified and supports only one physical channel for R2D and D2R, respectively.

The physical channel processing flow is shown in Figure 5(a).

- Coding is separately considered for R2D and D2R. For R2D, line coding is employed to combat large SFO at the device, timing is maintained by detecting transition edges. No FEC is used for R2D considering decoding complexity for device. For D2R, since the reader is more advanced for complicated detection and processing, FEC and repetition are applied to enhance reliability.
- D2R SFS is an important tool to support FDMA and to improve access efficiency.
- Cyclic redundancy check (CRC) can be attached to both links for reliability. Additionally, CRC can be omitted to save overhead in case of some extreme short messages with frequent transmissions.

Figure 5(b) depicts the frame structure. As an asynchronous system, the frame structure design for both links should support functionalities of start transmission indication and timing acquisition, which is achieved by preamble. For R2D, the preamble (i.e., time acquisition signal) considers the low-power detection of A-IoT devices, so it includes an additional start indicator part. A postamble can be used to indicate transmission termination, but this function can also be realized by R2D control information. The necessity of a postamble is still controversial. For D2R, considering the lower receiver sensitivity at the reader side, it is beneficial to introduce midamble and

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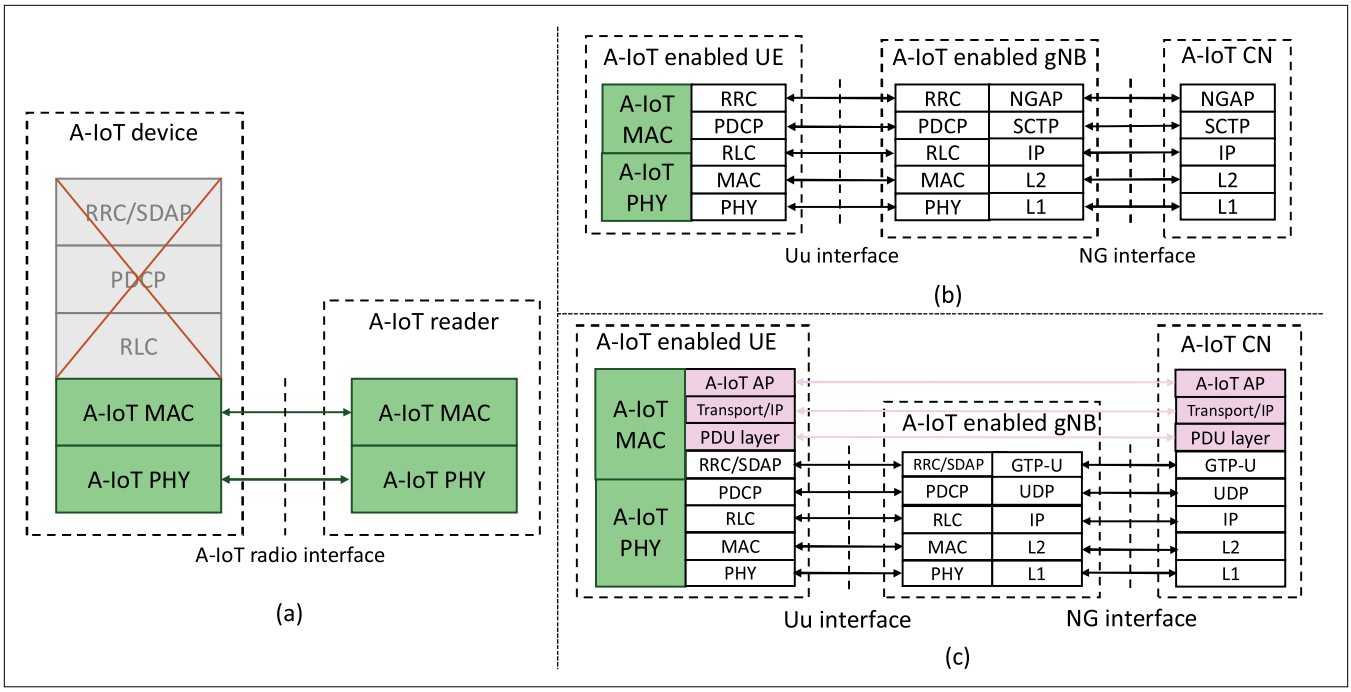


Figure 6. a) Protocol stack for A-IoT radio interface. b) Protocol stack for RRC-based solution for topology 2. c) Protocol stack for UP-based solution for topology 2.

postamble to assist SFO/CFO estimation, channel and interference estimation.

General Access Stratum Protocol Stack

For both topology 1 and 2, the next generation (NG) interface is applied between A-IoT radio interface and core network (CN) with necessary adaptations. In this section, protocol stack for A-IoT radio interface is discussed with special considerations on NR air interface for topology 2.

Protocol Stack for A-IoT Radio Interface

Topology is transparent to devices, and hence the protocol stack for A-IoT radio interface for both topologies are the same. The A-IoT radio interface is defined between a device and A-IoT enabled gNB (hereinafter called BS) for topology 1 and between a device and A-IoT enabled UE (hereinafter called intermediate UE) for topology 2. Due to low power consumption and complexity of a device, it is unreasonable to support heavy functionalities, such as, Abstract Syntax Notation One format, Quality of Service assurance, ciphering/deciphering, integrity protection, Automatic Repeat-reQuest, etc., and hence the Radio Resource Control (RRC) layer, Service Data Adaptation Protocol layer, Packet Data Convergence Protocol layer, and Radio Link Control layer are not required. An extremely simplified protocol stack is pursued and the only necessary layers for A-IoT radio interface are A-IoT physical (PHY) layer and MAC layer, as depicted in Figure 6(a). The A-IoT MAC layer supports necessary layer 2 functionalities including paging, random access, data transmission, and packet segmentation.

Protocol Stack for Topology 2

For topology 2, the air interface between intermediate UE and BS is additionally designed, and NR air interface is largely reused. Furthermore, regarding the protocol layer used to carry the A-IoT messages in the NR air interface, two candidates, which are RRC-based and User Plane (UP)-based solutions, are considered, as depicted in Figure 6(b) and (c). For RRC-based solution, the BS packages the service from CN into an RRC message and sends it the intermediate UE. The BS can also perform radio resource management as it is aware of the service request. Both CN and BS can perform intermediate UE selection. On the other hand, for UP-based solution, a PDU session is established between intermediate UE and UP function to transfer the A-IoT service. As the service request is transparent to the BS, assistance information from CN or intermediate UE may be required to manage A-IoT radio resources. Only CN can perform intermediate UE selection.

Access Procedures

For inventory and command use cases, the overall access procedure is formulated as 3 steps [5]:

- Step A: The reader transmits a paging message to trigger one or a group of or all device(s) to respond. The paging message may include inventory and/or command messages.
- Step B: The triggered device(s) performs the random access procedure in response to paging and transmits the device identifier (ID) to the reader.
- Step C: Subsequent R2D and D2R data transmission is performed to send command and corresponding response.

Common and Dedicated A-IoT Paging (Step A)

A paging message carries one or multiple device ID(s) corresponding to one, a group of or all devices. If the paging message targets all or a group of devices, contention-based random access (CBRA) is applied. Otherwise, if it targets one or multiple known devices associated with

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dedicated access resources, the device(s) can simply perform contention-free random access.

Random Access Procedure (Step B)

3-step and 2-step CBRA are considered. The 3-step CBRA procedure includes:

- Message 1: Device transmits 16-bit random ID.
- Message 2: Reader transmits contention resolution containing the random ID successfully received by reader. Contention resolution succeeds if Message 2 carries the same random ID as the device transmitted in Message 1.
- Message 3: Device transmits device ID to reader.
- Feedback (Optional): Reader may transmit the feedback to indicate whether Message 3 is successfully received.

2-step CBRA combines Message 1 and Message 3. The device ID is sent to the reader along with the random ID in Message 1. The two solutions are applicable to different conditions. For a small number of devices, 2-step CBRA is more latency efficient. For large number of devices, 3-step CBRA uses a shorter Message 1, which is less resource-intensive and provides more flexibility in contention resolution, thus outperforming 2-step CBRA.

Data Transmission Scheduling (Step C)

A-IoT data transmission utilizes a short Access Stratum (AS) ID because of two reasons. First, with the limited device complexity, it is impossible to support Cell-Radio Network Temporary ID scrambled scheduling. Besides, applying the full device ID for AS scheduling is too power consuming. The reader either reuses the random ID in Message 1 as the AS ID or assigns a new AS ID in Message 2 when that in Message 1 is occupied by other devices in the same paging round.

In addition, the D2R transmission scheduling is related to the message size transmitted by the device, which is beneficial to be known by the reader for efficient scheduling. The message size can be reported to the reader by the device. On the other hand, if the message size is available at CN, the CN can provide the size to the reader.

Moreover, transmission failure handling may be needed to improve scheduling reliability. The reader can send explicit failure or success indication to inform whether the device needs to re-access. Re-access occasions can also be provided by such indication. The device ID in the paging message can be used to determine whether this paging is for re-access or for new access.

Outlook

The upcoming 3GPP A-IoT normative work in Rel-19 concentrates on indoor scenario and device 1. It is anticipated that future evolution to outdoor solutions targeting device 2 or more advanced devices will significantly enhance coverage, capacity, and data rates, among other performance metrics. Although topology 2 is not specified in Rel-19, the inclusion in subsequent releases, such as Rel-20, holds the potential to broaden the scope of A-IoT applications. We eagerly anticipate the deployment of A-IoT solutions on the operators' licensed spectrum, which introduces new capabilities and contributes to a market enhancement in user experience.

Conclusion

In this paper, we provided the state-of-the-art overview of 3GPP A-IoT radio access aspects, and thoroughly discussed aspects that make A-IoT outperform the existing passive IoT technologies, incorporating various evaluation results. In terms of link performance, 10% BLER is achieved at -4.5 dB via techniques such as precise SFO estimation and FEC, significantly extending the D2R coverage. Inventory efficiency improves by more than 50% via FDMA and hybrid bi-tree algorithm. Energy harvesting solutions and packet segmentation extends the device availability time. Coexistence is guaranteed from R2D waveform design. Depending on the promising features, A-IoT solution will be specified in 3GPP Rel-19. In Rel-20, A-IoT designs toward novel scenario and device type will be further pursued.

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